

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460688.802 +/- 0.011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.106 +/- 0.042
Transit depth (Rp/Rs)^2	1.12 +/- 0.89 [%]
Orbital Inclination (inc)	87.18 +/- 2.49
Airmass coefficient 1 (a1)	0.933 +/- 0.021
Airmass coefficient 2 (a2)	0.06 +/- 0.33
Scatter in the residuals of the lightcurve fit is	2.22 %
Best Comparison Star	#3 - [595, 416]
Optimal Method	PSF photometry
Transit Duration (day)	0.055 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.106 $\pm$ 0.042 vs 0.11 (deviation 0.08 σ)
Duration fit vs published	0.055 d vs 0.1208 d (deviation 2.62 σ)
Mid-time O-C	23.5 min
Depth SNR	2.1
Residual scatter	2.22 %

Light curve

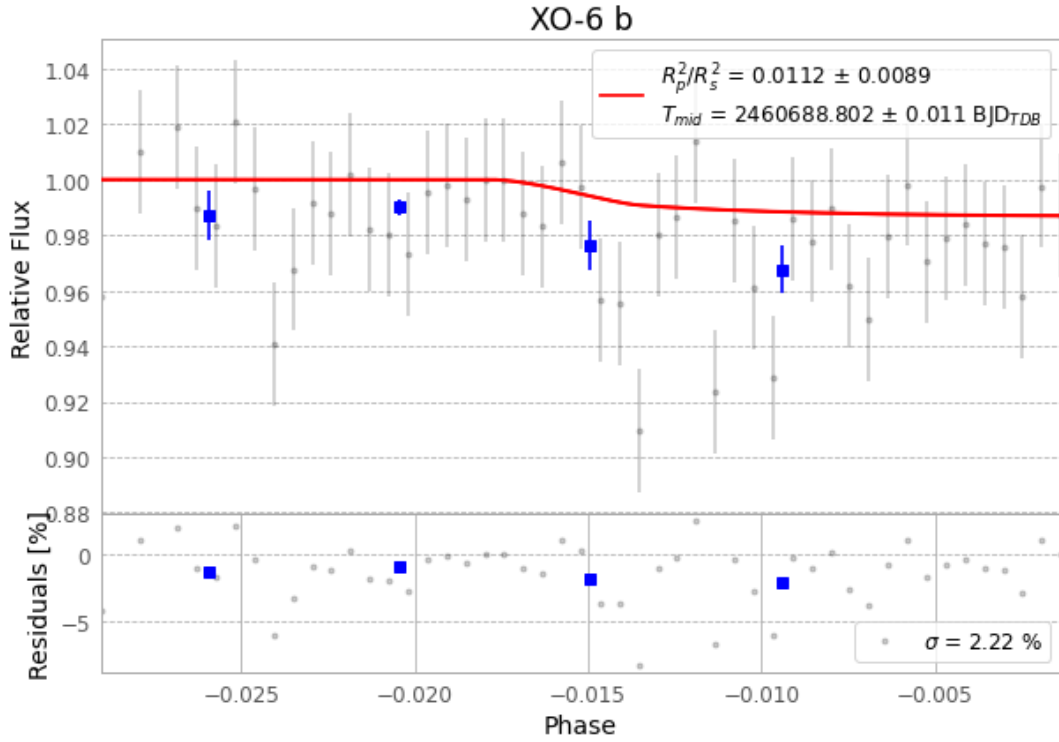


Figure 1 — FinalLightCurve_XO-6 b_2025-01-12.png (SHA-256 6f5377fe376ca723...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [463, 299], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 b6f713b2285fc017...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

49 FITS frames (32 MB), XO-6250113043018.FITS → XO-6250113070016.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-6-250113.log (0ffc275b4ddc...), FinalLightCurve_XO-6 b_2025-01-12.png (6f5377fe376c...), FinalLightCurve_XO-6 b_2025-01-12.csv (38c947699a7a...)

Compute resources

Wall time 92.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 23:54Z.